

Multicalibration Is Necessary for Unbiased Model-Based Prevalence Estimation

Significance

Researchers increasingly use AI systems—including large language models—as measurement devices to estimate how common a phenomenon is in a population. A critical but overlooked problem arises when the target population differs from the one on which the device was validated: standard calibration methods produce biased prevalence estimates. We show that multicalibration—requiring a device to be accurate not just on average but across all relevant subgroups—is the necessary and sufficient condition for unbiased prevalence estimation under population shift. Our result, confirmed by simulation and empirical application to U.S. employment data, implies that the rapidly growing body of research using AI for measurement is vulnerable to systematic bias unless multicalibration is enforced.

Abstract

Estimating the prevalence of a category in a population using imperfect measurement devices—diagnostic tests, classifiers, or large language models—is fundamental to science, public health, and platform governance. Standard approaches correct for known device error rates, but assume these rates remain stable across populations. We show this assumption fails under covariate shift: when the composition of the target population differs from the calibration population, globally calibrated devices produce biased prevalence estimates. We prove that multicalibration—calibration conditional on all relevant subgroups—is both necessary and sufficient for unbiased prevalence estimation under arbitrary covariate shift, connecting recent theory from algorithmic fairness to a longstanding problem in measurement. A simulation study demonstrates that standard methods (Classify & Count, Rogan-Gladen adjustment, isotonic regression) exhibit bias growing with the magnitude of distribution shift, while a multicalibrated estimator maintains near-zero bias throughout. In an empirical application using the American Community Survey, we estimate employment prevalence under synthetic age distribution shifts across 14 U.S. states. Standard methods show bias of 2–19 percentage points; a multicalibrated predictor reduces bias to under

1.2 percentage points in all scenarios. These results have immediate implications for the expanding use of AI systems as zero-shot measurement devices, where extrapolation to new populations is the primary purpose.

Introduction

Measuring the prevalence of a category within a target population is common across the sciences. Epidemiologists estimate disease prevalence from imperfect diagnostic tests (Rogan and Gladen 1978; Pollán et al. 2020; Neuhauser et al. 2022). Social scientists measure the frequency of political events, attitudes, or behaviors from coded text (Grimmer et al. 2022). Platform operators quantify rates of harmful content to assess policy effectiveness. In each case, the researcher relies on an imperfect measurement device—a test, classifier, or human coder—and must correct for its errors to recover the true population rate.

Large language models (LLMs) have dramatically expanded this enterprise. Researchers now deploy LLMs as zero-shot measurement devices to estimate prevalence at scales previously impossible. Applications span democracy indicators (Weidmann et al. 2025), protest events (Overös et al. 2024), survey response coding (Mellon et al. 2024; Gilardi et al. 2023), clinical attribute extraction (Sushil et al. 2024; Sibley et al. 2025), art classification (Tojima and Yoshida 2025), and multilingual content analysis (Karjus et al. 2025; Halterman 2025). The central promise is extrapolation: estimating prevalence in new populations, time periods, or demographic subgroups without additional annotation. Yet virtually none of these studies address what happens to prevalence estimates when the target population differs from the one on which the device was validated.

Standard correction methods assume that device error rates (sensitivity, specificity, or calibration curves) remain constant across populations (González et al. 2017). This assumption fails whenever the target population has a different composition of features—a condition called *covariate shift* or *mix shift*—which is precisely the setting where extrapolation is most needed. We show that the resolution lies in *multicalibration*: a device must be calibrated not just on average, but simultaneously across all relevant subgroups. This requirement, introduced in the algorithmic fairness literature (Hébert-Johnson et al. 2018), turns out to be both necessary and sufficient for unbiased prevalence estimation under arbitrary covariate shift, a connection established theoretically by Kim et al. (2022).

Bias, rather than variance, is the central concern. In prevalence estimation, variance decreases with sample size and is typically small when devices are applied to large corpora or surveys. Bias, however, is a systematic error that persists regardless of sample size. A biased prevalence estimate does not become more accurate with more data—it converges to the wrong answer. This makes the conditions under which bias arises, and how to eliminate it, the primary methodological question.

We make three contributions. First, we demonstrate that all standard prevalence estimation methods—Classify & Count, Adjusted Count (Rogan-Gladen), and global calibration—fail under covariate shift, and explain *why* through a unified framework. Second, we show that multicalibration provides a constructive solution: practical algorithms exist to post-hoc multicalibrate any predictor, yielding a device whose prevalence estimates are robust to population shift. Third, we validate these results empirically using the American Community Survey, showing that a multicalibrated predictor achieves near-zero bias under substantial age distribution shifts where standard methods fail by up to 19 percentage points.

Results

Standard calibration methods fail under covariate shift

Consider a binary outcome $Y \in \{0, 1\}$, features X , and a device $h(X) \in [0, 1]$ producing probabilistic predictions. The goal is to estimate the population prevalence $\pi = P(Y = 1)$ in a target population using only the device’s outputs. We assume that the causal direction is $X \rightarrow Y$, so that $P(Y|X)$ is stable across populations while the marginal $P(X)$ may shift—the standard covariate shift assumption (Wu and Resnick 2024b). This is the typical setting for measurement: features (demographics, text content, patient characteristics) causally influence outcomes (employment, attitudes, diagnoses), and applying a device to a new population changes the mix of features but not the underlying feature-outcome relationships.

A device is *globally calibrated* if $\mathbb{E}[Y \mid h(X) = p] = p$ for all prediction values p . Under global calibration, $\mathbb{E}[h(X)] = \mathbb{E}[Y] = \pi$, so the sample mean of predictions is an unbiased prevalence estimate. However, global calibration is a property of a *specific* population. A device calibrated on one population need not be calibrated on another, even under covariate shift with stable $P(Y|X)$.

The failure mechanism is straightforward. Suppose the population consists of subgroups indexed by g , with sizes w_g and group-specific calibration errors $\epsilon_g = \mathbb{E}[h(X) - Y \mid G = g]$. Global calibration requires only that these errors cancel on average: $\sum_g w_g \epsilon_g = 0$. Under covariate shift, the weights change to w_g^* , and the bias in the prevalence estimate becomes $\sum_g w_g^* \epsilon_g$, which is generally nonzero unless every $\epsilon_g = 0$. Crucially, a device can be perfectly globally calibrated—with errors that precisely cancel in the training population—while being substantially miscalibrated on every individual subgroup.

This failure applies to all standard methods. *Classify & Count* binarizes predictions at a threshold τ and reports the positive fraction; the threshold is chosen to match true prevalence on calibration data, but this correspondence breaks under shift. The *Adjusted Count* (Rogan-Gladen) estimator $\hat{\pi} = (\hat{p} -$

$\text{FPR}/(\text{TPR} - \text{FPR})$ uses sensitivity and specificity from calibration data (Rogan and Gladen 1978). These error rates are weighted averages of group-specific rates; when group proportions shift, the learned rates no longer match the target population’s. The ratio estimator further amplifies errors when $\text{TPR} - \text{FPR}$ is small. The same argument applies to calibration approaches using stable error rates such as the Calibrate-Extrapolate framework (Wu and Resnick 2024a, 2024b). *Isotonic regression* and other global calibration methods learn a single mapping from scores to probabilities, which reflects the calibration population’s composition and need not hold elsewhere.

Simulation. We illustrate these failures with a controlled simulation. We generate data with a binary covariate $X \in \{0, 1\}$ and binary outcome Y , where $P(Y = 1|X = 1) = 0.85$ and $P(Y = 1|X = 0) = 0.15$. A classifier produces systematically biased predictions: 10% underestimation when $X = 0$ and 10% overestimation when $X = 1$. All calibration parameters are learned on a balanced training distribution ($P(X = 0) = 0.5$). We then evaluate prevalence estimates on test distributions where $P(X = 0)$ ranges from 0.01 to 0.99, repeating 50 times (details in Materials and Methods).

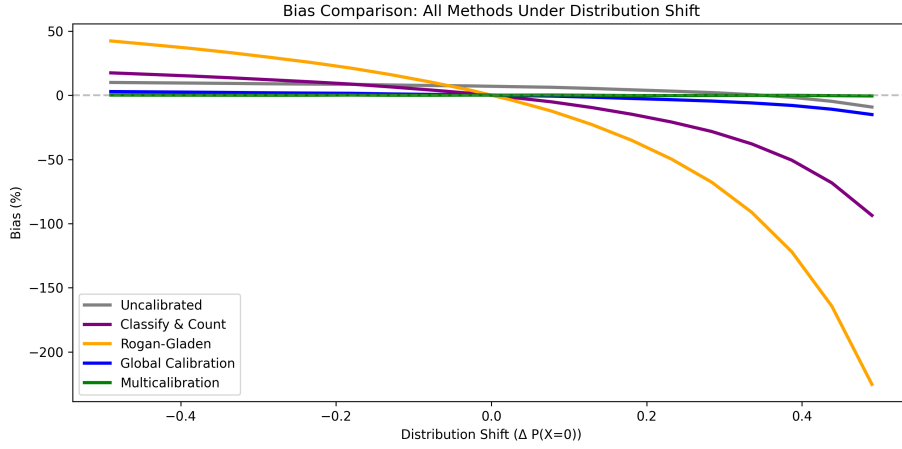


Figure 1: Figure 1

Figure 1: Average bias (%) of prevalence estimates under covariate shift for five methods across 50 simulation runs. Only multicalibration maintains near-zero bias across all levels of distribution shift.

Figure 1 shows the results. At the training distribution (center), all methods except the uncalibrated baseline produce approximately unbiased estimates. As the distribution shifts, the methods diverge. Classify & Count and Rogan-Gladen show rapidly growing bias, exceeding $\pm 40\%$ at extreme shifts. The Rogan-Gladen estimator is particularly unstable: its ratio structure amplifies estimation errors, producing bias that exceeds the other methods by an order of magnitude. Global calibration shows more moderate but still substantial bias

(up to $\pm 15\%$). The uncalibrated baseline, already biased at the center due to the systematic miscalibration, shows additional shift-dependent bias. In contrast, the multicalibrated estimator maintains near-zero bias across the entire range of distribution shifts.

Multicalibration guarantees unbiased prevalence estimation

The simulation motivates the question: what property of a device is *sufficient* to guarantee unbiased prevalence estimation under arbitrary covariate shift? The answer is multicalibration.

A predictor $f(X)$ is α -*multicalibrated* with respect to a collection of subgroups $\mathcal{G}$ if, for every group $G \in \mathcal{G}$ and every prediction value v :

$$|\mathbb{E}[Y \mid f(X) = v, X \in G] - v| \leq \alpha,$$

whenever the conditioning event has sufficient probability mass (Hébert-Johnson et al. 2018). This strengthens global calibration by requiring accuracy not just on average, but within each subgroup at each score level.

The connection to prevalence estimation is direct. If f is multicalibrated with respect to $\mathcal{G}$, then $\mathbb{E}[f(X) \mid X \in G] \approx \mathbb{E}[Y \mid X \in G]$ for every $G \in \mathcal{G}$. Under covariate shift—where only $P(X)$ changes while $P(Y|X)$ remains stable—the law of iterated expectations gives:

$$\mathbb{E}^*[f(X)] = \sum_g w_g^* \cdot \mathbb{E}[f(X) \mid G = g] \approx \sum_g w_g^* \cdot \mathbb{E}[Y \mid G = g] = \mathbb{E}^*[Y] = \pi^*.$$

Because multicalibration ensures predictions are correct *within each subgroup*, the average of predictions tracks the true prevalence under *any* reweighting of those subgroups. This is the “Universal Adaptability” result of Kim et al. (2022): a multicalibrated predictor competes with propensity scoring for arbitrary target distributions, without requiring knowledge of the shift.

Returning to the framework from the previous section: global calibration only ensures $\sum_g w_g \epsilon_g = 0$ for the training weights w_g . Multicalibration ensures $\epsilon_g \approx 0$ for each subgroup individually, making the estimate robust to arbitrary changes in w_g . This is also a *necessary* condition: if any $\epsilon_g \neq 0$, there exists a target distribution (one that overweights group g) under which the prevalence estimate is biased.

The requirement is not merely theoretical. Practical post-hoc multicalibration algorithms exist (Hébert-Johnson et al. 2018; Gopalan et al. 2022) and can be

applied to any base predictor—including LLMs and black-box classifiers—by iteratively correcting predictions on subgroups where they are miscalibrated. The simulation in Figure 1 confirms this: the multicalibrated estimator, which learns additive corrections for each stratum of X , achieves near-zero bias regardless of the target distribution’s composition.

Empirical application: employment prevalence under age distribution shift

We validate these results using the American Community Survey (ACS), a large-scale annual survey conducted by the U.S. Census Bureau. The prediction task is binary employment status, with 16 sociodemographic features including age, education, marital status, disability status, and citizenship.

Setup. We train a logistic regression classifier on data from eight U.S. states (TX, MI, PA, OH, IL, GA, NC, VA) across 2016–2018, totaling approximately 1.5 million training observations. The remaining in-distribution data is split into a calibration set ($n \approx 644,000$) and a test set ($n \approx 920,000$). Six additional states (CA, NY, FL, WA, AZ, CO) are held out for out-of-distribution (OOD) evaluation. We fit two post-hoc calibration methods on the calibration set: isotonic regression (global calibration) and MCGrad, a multicalibration algorithm that enforces calibration conditional on subgroups defined by both categorical and numerical features. We compare five prevalence estimation methods: raw (uncalibrated) scores, Classify & Count with a prevalence-matched threshold, Rogan-Gladen adjustment, isotonic regression, and MCGrad.

Age distribution shift. Employment rates vary dramatically by age: approximately 47% for ages 16–24, 76% for ages 25–54, 61% for ages 55–64, and 17% for ages 65+. This makes age an ideal dimension along which to construct meaningful distribution shifts. We create synthetic target populations by re-sampling test data with shifted age distributions: young-skewed (oversampling ages 16–30), old-skewed (oversampling ages 60+), and bimodal (oversampling both tails). The resulting populations have true employment rates ranging from 12.8% to 46.0%. All calibration parameters are estimated once on the original calibration set and held fixed across scenarios.

Figure 2: Prevalence estimation bias (percentage points) under synthetic age distribution shift, for in-distribution data (left) and out-of-distribution states (right).

Setting	Age Dist.	True Prev.	Raw Scores	CC	Rogan-Gladen	Isotonic	MCGrad
In-Dist	Original	46.0%	-0.31	-0.07	+0.26	-0.30	-0.27
In-Dist	Young-skewed	12.8%	+1.93	+2.47	-12.82	+2.04	-0.11

Setting	Age Dist.	True Prev.	Raw Scores	CC	Rogan- Gladen	Isotonic	MCGrad
In-Dist	Old-skewed	16.8%	+7.23	-6.62	-16.77	+6.65	+0.22
In-Dist	Bimodal	21.1%	+4.57	-1.50	-18.62	+4.33	+0.12
OOD	Original	45.1%	+1.15	+1.40	+2.12	+1.17	+1.35
OOD	Young-skewed	13.0%	+2.93	+3.73	-12.96	+3.08	+0.88
OOD	Old-skewed	16.0%	+8.47	-5.64	-15.97	+7.91	+1.01
OOD	Bimodal	20.8%	+5.91	+0.09	-16.14	+5.69	+1.13

Table 1: Prevalence estimation bias in percentage points (pp) under synthetic age distribution shift.

Results. With no age shift, all methods produce approximately unbiased estimates (Table 1). Under shift, the methods diverge sharply.

Rogan-Gladen exhibits catastrophic failure, with bias of -12.8 to -18.6 percentage points in the in-distribution setting—effectively estimating zero or negative prevalence when the true rate is 13–21%. This occurs because TPR and FPR, estimated on the original age distribution, become invalid when the age mix shifts. Isotonic regression and raw scores show substantial bias of 2–8 percentage points: their global calibration curve, accurate on average in the training distribution, no longer reflects the true relationship when particular age groups are over- or under-represented. Classify & Count shows bias of up to 6.6 percentage points as the learned threshold loses its validity.

MCGrad produces near-zero bias across all in-distribution age-shift scenarios (≤ 0.27 pp). Because MCGrad is multicalibrated with respect to age and other features, its predictions are correct *conditional on* these covariates. When the age distribution shifts, the subgroup-level predictions remain valid, and their average tracks the true prevalence.

In the OOD setting—applying the model to states never seen during training *and* shifting the age distribution—MCGrad’s advantage persists with modestly larger bias (0.88–1.35pp), reflecting the additional geographic shift along an axis not included in the calibration subgroups. Even so, MCGrad’s bias remains 3–8 $\times$ smaller than the next best method.

Discussion

We have shown that multicalibration—calibration conditional on subgroups, not just on average—is the key property for accurate model-based prevalence

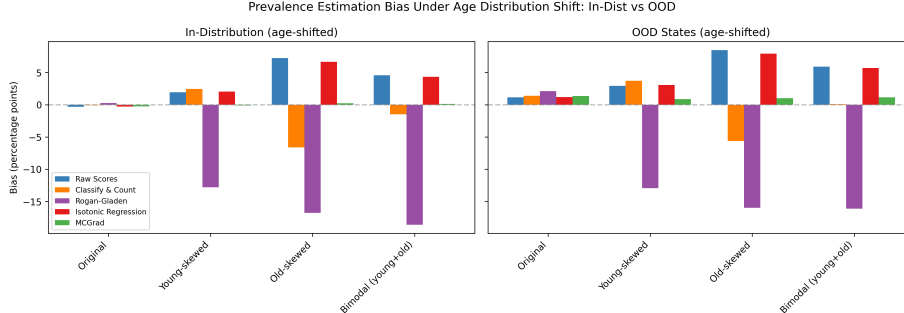


Figure 2: Figure 2

estimation under population shift. Standard calibration methods can produce severely biased estimates when the composition of the target population differs from the calibration population. Multicalibration resolves this by ensuring accuracy at the subgroup level, making prevalence estimates robust to arbitrary reweightings.

The practical implications are immediate. The growing use of LLMs as zero-shot measurement devices rests on the assumption that models validated on one population can be reliably applied to others. Our analysis shows this assumption is safe only when the model is multicalibrated with respect to features that may shift across populations. Without this, prevalence estimates in new populations—the primary use case for LLM-based measurement—are unreliable. Researchers using AI-based measurement should treat multicalibration not as an optional fairness criterion, but as a methodological requirement for valid inference.

This result also clarifies why certain widely-used practices are inadequate. The common approach of reporting a model’s overall accuracy, F1, or AUC on a held-out set provides no information about calibration, let alone multicalibration. A model with 95% accuracy can still produce heavily biased prevalence estimates if its errors are unevenly distributed across subgroups. Similarly, reporting a single calibration curve—as is standard in the quantification literature (González et al. 2017)—certifies only global calibration, which we have shown is insufficient under shift. Even comprehensive methodology textbooks for text-as-data research devote extensive attention to classification accuracy and validation but do not discuss calibration as a requirement for valid prevalence estimates (Grimmer et al. 2022); multicalibration is entirely absent from these discussions.

Several limitations warrant discussion. First, multicalibration requires specifying the collection of subgroups $\mathcal{G}$. The guarantee holds only for shifts along axes represented in $\mathcal{G}$; if the relevant dimension of shift is absent, bias can persist—as reflected in the modest OOD bias in our empirical results, where geographic

shift introduces variation beyond the calibrated features. In practice, $\mathcal{G}$ should include all features plausibly associated with both the outcome and the shift. Second, multicalibration requires labeled calibration data of sufficient size to estimate subgroup-specific corrections. This is feasible for standard ML pipelines but more challenging for LLMs used in truly zero-shot settings, where obtaining calibration labels may require the manual annotation that LLMs were intended to avoid. Prediction-powered inference (Angelopoulos et al. 2023) offers a complementary framework for combining small labeled datasets with large-scale LLM predictions. Third, modern LLMs produce text rather than probability scores; calibratable scores can be obtained via token log-probabilities, confidence-elicitation prompts, or repeated sampling, but developing practical multicalibration pipelines for these outputs remains an open problem. Fourth, our framework assumes covariate shift ($P(Y|X)$ stable). Under concept drift, where the feature-outcome relationship itself changes, no purely statistical correction can substitute for new labeled data from the target distribution. Finally, our empirical application uses synthetic distribution shifts to evaluate bias. This design choice is deliberate: measuring bias requires knowing the true prevalence in the target population, which is generally unavailable under naturally occurring shifts. The synthetic shifts we construct are realistic in magnitude and correspond to substantively meaningful population differences.

These results connect two literatures that have developed in isolation. The quantification literature has focused on correcting aggregate error rates but has not engaged with subgroup-level calibration (González et al. 2017; Wu and Resnick 2024b). The multicalibration literature has focused on individual-level prediction quality and fairness but has not emphasized the implications for population-level inference (Hébert-Johnson et al. 2018; Kim et al. 2022). Our contribution is to show that subgroup-level calibration is the missing link that makes model-based prevalence estimation reliable under the distribution shifts that motivate its use.

Materials and Methods

Simulation. We generate $n = 10,000$ observations with a binary covariate $X \in \{0, 1\}$ and outcome $Y \sim \text{Bernoulli}(P(Y|X))$, where $P(Y = 1|X = 1) = 0.85$ and $P(Y = 1|X = 0) = 0.15$. The classifier produces deterministic scores $\hat{p}(X = 0) = 0.135$ and $\hat{p}(X = 1) = 0.935$, representing 10% multiplicative bias within each stratum. For each of $B = 50$ iterations, we generate fresh calibration data from $P(X = 0) = 0.5$, estimate all method-specific parameters, then evaluate bias on 20 test distributions with $P(X = 0)$ ranging from 0.01 to 0.99. The five methods compared—uncalibrated averaging, Classify & Count with a prevalence-matched threshold, Rogan-Gladen adjustment, global multiplicative calibration, and multicalibration with stratum-specific additive corrections—are detailed with full mathematical definitions in the SI Appendix.

Empirical application. We use the ACS Public Use Microdata Sample via the *folktables* package, with the ACSEmployment prediction task (binary: employed vs. not employed) and 16 sociodemographic features. Training data comprises eight states (TX, MI, PA, OH, IL, GA, NC, VA) across 2016–2018. The base model is logistic regression with standard scaling. Post-hoc calibration uses isotonic regression (global) and MCGrad (multicalibration with categorical and numerical segment features) on a held-out calibration set ($n \approx 644,000$). Synthetic age-shifted populations are created by importance-weighted resampling of the test set ($n \approx 920,000$), with exponential weights favoring young ages, old ages, or both extremes. Six additional states (CA, NY, FL, WA, AZ, CO) serve as OOD evaluation data.

Software. MCGrad is available at github.com/facebookincubator/MCGrad. Simulation and analysis code are available at [repository URL].

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